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Concentrated and piped sunlight for indoor illumination

L. M. Fraas, W. R. Pyle, and P. R. Ryason

A concept for indoor illumination of buildings using sunlight is described. For this system, a tracking concentrator on the building roof follows the sun and focuses sunlight into a lightguide. A system of transparent lightguides distributes the sunlight to interior rooms. Recent advances in the transparency of acrylic plastic optical fibers suggest that acrylic lightguides could be successfully used for piping sunlight. The proposed system displaces electricity currently used for indoor lighting. It is argued that using sunlight directly for indoor illumination would be about twenty-five times more cost-effective than using sunlight to generate electricity with solar cells for powering electric lamps for indoor lighting.

I. Introduction

It is well known that solar energy can be converted to heat or electrical energy and that these forms have value. It is also well known that the sun's energy can be used for illumination. Although the value of sunlight is not generally appreciated, the fact that sunlight is the most valuable form of solar energy has been pointed out by various authors.¹⁻⁴

To appreciate the value of sunlight, imagine two alternative solar energy systems: In the first system, solar cells capture sunlight at a desert utility site and convert it to electricity with an efficiency of 10%. Then the electricity is routed through electrical cables to a building. In the building, the electricity is converted back to light with an efficiency of 20%.¹ For this system, only 2% of the solar energy is delivered as useful illumination energy. For the second system, imagine that the sunlight is captured on the building roof and concentrated and routed with optical cables to overhead lamp fixtures with only 50% transmission loss. This second system delivers the solar energy with an overall efficiency of 50%. The sunlight is used twenty-five times more efficiently in the second system. Since a solar illumination system can displace electricity, the energy in sunlight has more value as light than as electricity.

There are some subtle points in the arguments just made. It was assumed that solar indoor illumination can displace electricity by substituting for electric

lighting. Therefore, we exclude from consideration sunlight from windows and skylights since these forms of solar lighting have been used for centuries. Although physiological and psychological health problems have been related to the use of artificial lighting,⁵⁻⁷ electric lights have some special and very desirable qualities not available from window lighting. Some of these features are: (1) overhead illumination, (2) illumination well into the interior of the building, (3) adequate illumination levels, (4) constant illumination levels, (5) illumination control (on/off, high/low, portability).

Window lighting does not allow for overhead illumination or illumination well into the interior of the building. Furthermore, the illumination level supplied from a window can vary by orders of magnitude throughout a day. For example, the direct sunbeam entering a window facing east in the morning is much more intense than the diffuse skylight entering the same window in the afternoon.

These problems with window lighting are rectified by the concept shown in Fig. 1 in which direct light from the sun is collected by a tracking concentrator and focused into a lightguide for distribution into overhead lighting fixtures in the building interior. The idea of a tracking concentrator for indoor illumination has been described earlier by Duguay and Edgar.¹ In the southwestern United States, where the sun is seldom obscured by clouds, this tracking scheme provides a bright and constant light level at the concentrator focus throughout the day (the focused light is an image of the sun). However, the scheme of Duguay and Edgar¹ did not use lightguides but rather a system of lenses and mirrors to route the light beam. A lightguide would be preferable, in that one would avoid problems of optical alignment and the necessity of specially designed buildings. One could install lightguides in a manner analogous to installing a fire-extinguishing sprinkler system or an electrical conduit distribution system.

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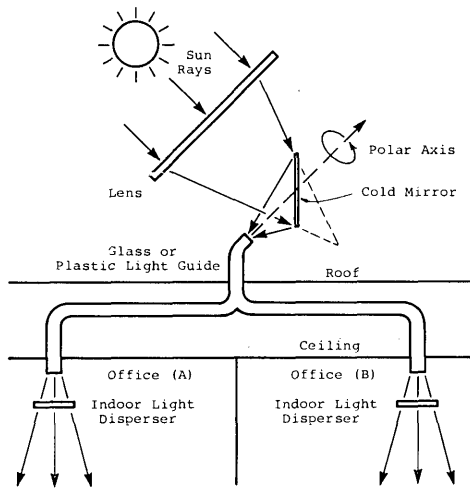


Fig. 1. Basic concept for solar illumination.

In Sec. II, a concentrated and guided sunlighting system is described in more detail. In Sec. III, the economic value of such a system is discussed. It is assumed that a typical fluorescent lighting system will be required for backup at night and on cloudy days. Then the sunlighting system will be cost-effective if the electricity displaced can pay for the capital cost of the solar lighting system.

II. System Description

In this section we will describe a hypothetical design for a solar lighting system for a three-story office building. We will provide sufficient detail to answer the following questions:

- (1) What are a desirable concentrator lens size and tracking system?
- (2) What would be a reasonable concentration ratio and lightguide diameter?
- (3) What kind of material would one use for the lightguide? Is there a low-cost material with sufficient transparency?

Figure 2 shows our assumed office building layout. We assume 3.3- × 3.3-m offices and one central trunk lightguide at a common corner of four offices. The trunk lightguide will supply light to four distribution lightguides on each floor for the three floors. One tracking lens will supply light for twelve offices or 120 m² of office space.

What size lens will be necessary? An area of 1 m² of sunlight exposure at normal incidence contains enough light to illuminate 50 m² of indoor office space, assuming a 50% light coupling efficiency. A person with photographic experience can relate to this 50:1 area factor by noting that outdoor lighting is 100 times more intense than indoor lighting; i.e., it takes an $f/\text{No.}$ of 1.8 (or a flash) for indoor photos and an $f/\text{No.}$ of 18 for outdoor photos. The square of the ratio of $f/\text{Nos.}$ gives the light-intensity ratio. Therefore, it will take a 2.4-m² lens to supply illumination for the 120-m² office space. Figure 3 shows a schematic of the tracking solar concentrator envisioned for this application. Specifically, a large plastic Fresnel lens collects and focuses the light.

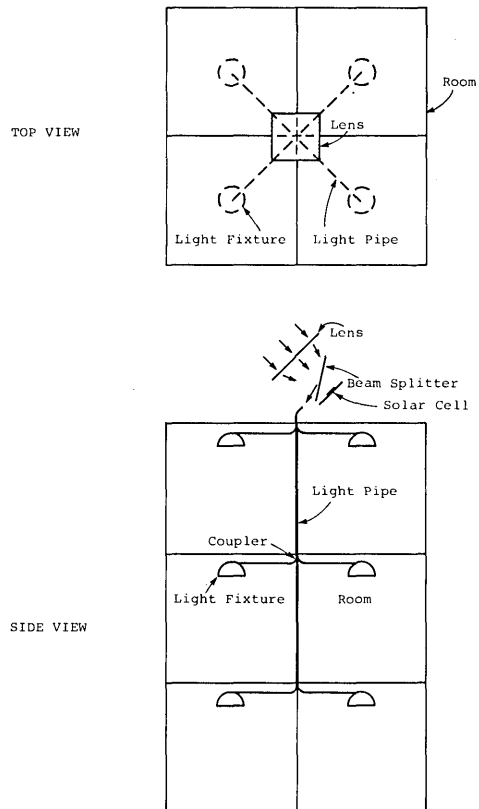


Fig. 2. Office building layout.

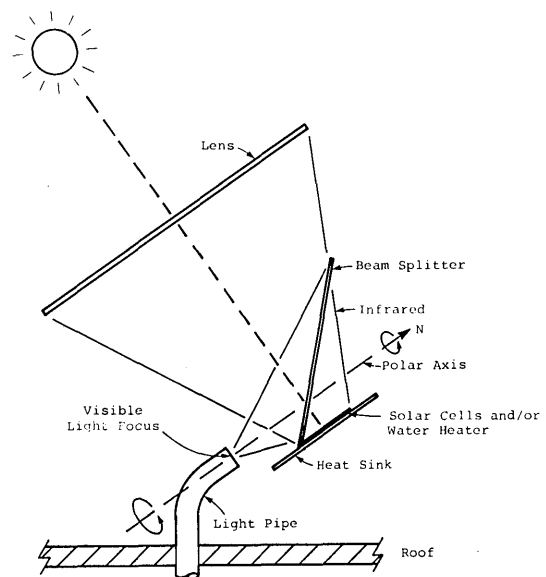


Fig. 3. Lens and tracker configuration.

The lens is tracked using polar-axis (north-south rotation-axis) tracking.

Converging light from this lens encounters a beam splitter, where it is reflected to a focus on the polar axis of rotation. At this point it enters a central light guide. The beam splitter serves to reflect the visible portion of the solar spectrum to the lightguide, and it transmits

the longer wavelength portion of the solar spectrum to solar cells for conversion to electricity [$\sim 50\%$ of sunlight is in the visible and ultraviolet, and 50% is in the infrared portion of the spectrum (1)]. From the point of view of a solar illumination lighting system, the beam splitter rejects infrared (heat). This rejected heat will reduce the air-conditioning load for the building and reduce heating of the lightguides. Returning to the light path, the fact that the lightguide entrance axis corresponds to the polar axis for lens tracking allows for a stationary lightguide in conjunction with the tracking Fresnel lens. The ratio of lens area to lightguide entrance area will be in the 100–400-range (100–400 suns) and typically 250. This would correspond to a tracking accuracy requirement of $\pm 2^\circ$ for an $f/1$ lens.⁸ This tracking requirement is easily achieved using clock motor gear drives. In fact, a polar-axis tracker of approximately this size has been built for remote concentrator photovoltaic applications. Models designed to operate with concentration ratios of 100 and 400 have been built by GE and RCA.^{9,10}

The central lightguide runs from the roof down into the building. In the case of Fig. 2, it is shown located at the common corner of four offices. It is envisioned that this central guide might have a circular cross section on the roof, with a graded section going to a rectangular cross section in the building. A rectangular cross section may accommodate branched distribution lines more easily than a circular cross section. The branching distribution lines are discussed in the next paragraph. This central lightguide can be constructed out of solid glass or plastic, or it might consist of a transparent tube filled with a transparent liquid.¹¹

This light pipe works by trapping all light entering it within a certain acceptance cone angle (see Fig. 4), which is characteristic of the transparent medium. For a transparent medium such as acrylic with a 1.5 index of refraction (n) surrounded by a lower-index cladding of a material such as Teflon ($n = 1.4$), this critical acceptance half-angle is 32.2° as shown in Fig. 4. A critical parameter describing this guide will be its optical loss for a given length. For a three-story building, as shown in Fig. 2, the longest light path will be ~ 15 m. Figure 5(a) shows the losses for a soda-lime glass pipe as a function of length.¹² The losses are $< 50\%$ in 15 m.

Recently, the losses for plastic optical fibers were significantly reduced by workers at Nippon Telephone and Telegraph (NTT).¹³ They removed particulates from the methyl methacrylate (MMA) monomer, the chain transfer agent, and the polymerization initiator by repeated distillation and washing. Vacuum distillation was used to minimize dissolved oxygen content. In this way, both the particulates and gas bubbles, which can scatter light, were minimized. Losses for NTT's acrylic fibers are shown in Fig. 5(b). Similar work was done by NTT with polystyrene core fibers.¹⁴

A 50%, or 3-dB, loss in 15 m equals a 200-dB loss/km. Acrylic fibers can meet this design requirement. In fact, plastic optical fibers have many advantages, in-

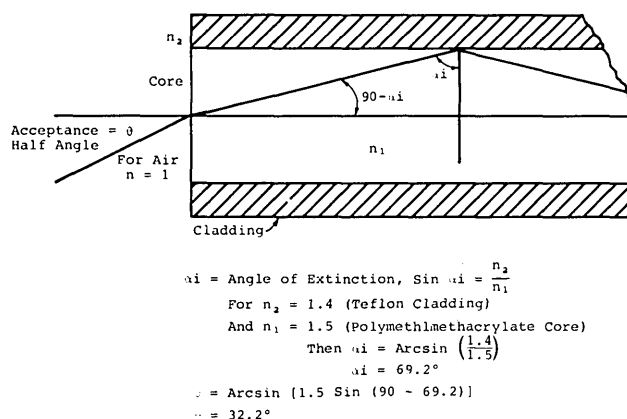
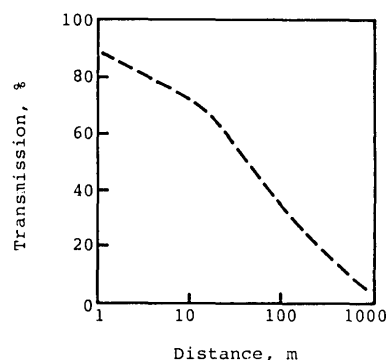
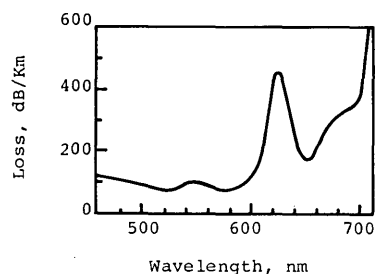


Fig. 4. Angle of total internal reflection.



(A)



(B)

Fig. 5. Transmission losses: (A) soda lime glass; (B) polymethylmethacrylate fiber.

cluding ease of splicing, good ductility (for making bends during the collection and distribution of sun-lighting), and a high numerical aperture in conjunction with a large core diameter (for easy coupling to light source). A lightguide for solar lighting would probably be a relatively large solid acrylic rod rather than a fiber bundle.

Since some of the loss in an optical fiber is due to absorption, it is of interest to estimate the temperature increase in a large solid acrylic rod transmitting the focused light of a lens into a building. For a lens of 2.4-m^2 area and a concentration ratio of 250, the diameter of the acrylic rod is 5.5 cm. Based on the loss data for plastic fibers^{13,14} and estimating that 32% of the

visible light energy entering the rod lies in the 600–750-nm wavelength range, ~30% of the loss (3 dB in 15 m) is due to absorption. The fractional loss/centimeter is 0.005, and a reasonable approximation is that this corresponds to a uniform rate of heat production in the large rod equal to 4×10^{-4} cal/cm³/sec. From the well-known expression for the exactly analogous problem of the temperature distribution in an electric wire carrying current,¹⁵ the temperature difference between the outside of the rod and its center is 13°C, a short distance from the entrance at steady state. This calculation assumes a uniform distribution of light across the face of the rod and that infrared radiation has been rejected by the beam splitter.

An alternative lightguide has been described, a transparent hollow pipe with prism-shaped outer walls.¹⁶ Light rays propagating in this guide undergo total internal reflection at the prism faces. Losses are higher than those for plastic fibers or for soda-lime glass, i.e., 8.2 dB in 15 m (85% loss). However, this form of lightguide may have utility in optimizing a system in which transmission loss is balanced against cost. The hollow pipe will use less material than a solid rod.

Sunlight could be routed from the central lightguide to the distribution guides via directional couplers with glued butt joints. The distribution guides with one or more large radius bends would run into light fixtures with reflecting sides and a translucent diffuser facing down into the room. Light could be turned on and off using simple shutters.

III. Economic Considerations

In this section, we first calculate the quantity and value of the electricity displaced by the system described above. Then we will make an estimate for the capital and installation cost for this system, thereby, arriving at a payback time for the system.

How much electricity does it take to light 120 m² of office space? In our laboratory, 320 W of fluorescent lighting are required to illuminate a 10-m² office. It is important to note that only 32 W of useful light is supplied downward into the office by the fluorescent lights. The remaining 288 W of the electrical power is rejected as heat. Therefore, the 2.4-m² lens in our hypothetical system would light twelve offices and displace 3.8 kW (12 * 320 W) of electrical power. Note that, since 1 m² of sunlight exposure contains ~1 kW of energy, the 2.4-m² lens collects 2.4 kW of energy but displaces 3.8 kW of electricity. How can we do that well? The answer is that a 6000 K light source (sun) contains more lumens (light)/watt than a fluorescent lamp does. So, 1 m² of light will displace 1.6 kW of electric power (3.8 kW/2.4 m²).

If the solar lighting system were installed in Phoenix, 1 m² of sunlight exposure would displace the 1.6 kW of electricity required for fluorescent lighting ~75% of the working time during normal working hours. This time corresponds to 1500 h/yr. Therefore, 1 m² of sunlight will displace 1.6 kW * 1500 h/yr or 1400 kW · h/yr. By analogy to photovoltaics, this electricity would have a premium value since it would be generated during times

Table I. Estimation of Lighting System Cost

Element	Description	Cost
Concentrator	2.5 m ²	\$750 (assuming \$300/m ²)
Central light-guide (material)	7.6 cm × 7.6 cm × 9.1 m	\$420 (assuming \$46/m)
Central light-guide (labor)	1.3 H/m	\$360
Distribution lightguides	12 lines, 6.4 cm ² by 1.5 m long	\$48 (assuming \$2.60/m)
Distribution (installation)	0.66 H/m, 18 m	\$360
Light fixtures	12 required	\$120 at \$10 each
Total		\$2058

of peak demand. Assuming a value of 8 cents/kW · h, 1 m² of sunlight is worth \$192/yr.

The above argument refers to the value of the electricity displaced by more efficient lighting. However, it is noteworthy that air-conditioning loads in the Pacific Southwest peak when the sun is brightest. For the fluorescent lighting system, the air-conditioning system will have to remove 1.6 kW/m² of heat, while for the solar lighting system, it will need to remove only 320 W of heat. (For 1 kW of solar radiation incident on the beam splitter, ~320 W is reflected as visible light in the spectral range between 4800 and 6200 Å.) The difference is 1.3 kW. For a coefficient of performance for the air-conditioning system of 2.5 (a typical value), the removal of 2.5 W of heat requires 1 W of electricity. Therefore, the solar lighting system saves an additional 520 W (780 kW · h/yr) of electricity/m² of collector aperture. This energy is worth \$63/yr (assuming 8 cents/kW · h). The total electric bill savings is then \$255/yr/m² of sunlight. It is appropriate to estimate the cost of the lighting system depicted in Fig. 2. The results are shown in Table I and indicate that the system merits further study. The costs considered include the concentrator and the lightguides (materials and installation). Data for these cost estimates were derived from Refs. 11–13. We assumed that acrylic would be used to construct the lightguides.

In summary, a system collecting 2.5 m² of sunlight and distributing that light to twelve offices in a three-story building will cost ~\$2100. The electricity it displaces will be worth 2.5 times \$255/yr or \$638/yr. This system will have a payback time of 3.2 yr.

The above considerations do not allow capital credits for (1) reduced size in the air-conditioning system or (2) solar tax credits.

IV. Conclusions

We have proposed a lighting system wherein sunlight can be used directly for indoor illumination to replace electric lighting.¹ This system has significant economic value. We have shown that concentrated sunlight has advantages in that the focused image of the sun is constant in intensity and bright enough to illuminate many offices simultaneously.² We have presented the design of a distribution system whereby concentrated light can be piped to various offices without complicated optical

alignment schemes and have shown that previously published¹³ transmission data for acrylic fibers suggest that acrylic can be used as a construction material for solar light pipes. Our economic evaluation suggests that such a concentrated and piped solar lighting system could prove economically viable in the southwestern United States at the present time.

The authors thank the referee for calling to their attention the paper on the hollow lightguide and for suggesting the effect of a high-power density on the lightguide be considered.

References

1. M. A. Duguay and R. M. Edgar, *Appl. Opt.* **16**, 1444 (1977).
2. M. A. Duguay, *Solar Electricity: The Hybrid System Approach*, *Am. Sci.* **65**, 422 (July-August 1979).
3. J. Parker, *Now We Have Solar Illumination*, *Mech. Illus.* **27**, (November 1977).
4. S. Selkowitz, *Effective Daylighting in Buildings (Parts 1 and 2)*, *Light. Des. Appl.* **9**, 6-8, 10-11, 61 (Feb. Mar. 1979).
5. J. N. Ott, *Health and Light* (Simon & Schuster, New York, 1976).
6. R. J. Wurtman, *The Effects of Light on the Human Body*, *Sci. Am.* **233**, 68 (1975).
7. L. Thorington, L. Parascandola, and L. Cunningham, *Visual and Biologic Aspects of an Artificial Sunlight Illuminant*, *J. Illum. Eng. Soc.*, **67**: 33-41, 1977.
8. R. O. Hughes, *Sol. Energy* **24**, 83 (1980).
9. N. F. Shepard, Jr., and T. S. Chan, *Fifteenth IEEE Photovoltaic Special Conf.*, p 336 (1981).
10. E. C. Boes, *Photovoltaic Systems Definition Project 1977 Annual Report*, SAND 78-0942, p. 57.
11. S. E. Miller *et al.*, *Proc. IEEE* **61**, 1703 (1973).
12. D. Kato and T. Nakamura, *J. Appl. Phys.* **47**, 4528 (1976).
13. T. Kaino, M. Fujika, S. Oikawa, and S. Nara, *Appl. Opt.* **20**, 2886 (1981).
14. T. Kaino, M. Fujii, and S. Nara, *Low Loss Polystyrene Core Optical Fibers*, *J. Appl. Phys.* **52**, 7061 (1981).
15. R. B. Bird, W. E. Stewart, and E. N. Lightfoot, *Transport Phenomena* (Wiley, New York, 1960), pp. 267-271.
16. L. A. Whitehead, R. A. Nodwell, and F. L. Curzon, *Appl. Opt.* **21**, 2755 (1982).

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